

Week 03: Inclusion-Exclusion Principle

Exercise 1. Answer the following questions:

- a) In a class, the number of students who are proficient in both C++ and Python is equal to the number of students who are not proficient in either. Given that there are 20 students proficient in C++ and 15 students proficient in Python, how many students are there in this class?
- b) How many positive integers n not exceeding 2025 satisfy $\gcd(n, 35) > 1$?

Exercise 2. Answer the following questions:

- a) In class 25C01, more than $\frac{1}{3}$ of the students are female and more than $\frac{1}{2}$ are members of the Youth Union. Additionally, more than $\frac{1}{6}$ of the male students are not members of the Youth Union. Prove that there is at least one female student who is a member of the Youth Union.
- b) There are 30 students in a dormitory: 15 students study painting, 8 study mathematics, and 6 study chemistry. Given that 3 students participate in all three courses, prove that there are at least 7 students who do not participate in any of these courses.

Exercise 3. How many ways are there to color a 4×4 grid with two colors, red or blue, such that no 3×3 subgrid is colored entirely in red?

Exercise 4. Similar to Exercise 3, how many ways are there to color the grid such that at least one of the following conditions is satisfied:

- The first row is entirely Red.
- The last column is entirely Blue.
- The 2×2 square at the top-left corner is entirely Red.

Exercise 5. A survey of 120 people regarding three news channels K_1, K_2, K_3 yielded the following results:

- 60 people watch K_1 , 50 people watch K_2 , 40 people watch K_3 .
- 25 people watch both K_1 and K_2 , 18 watch both K_1 and K_3 , 12 watch both K_2 and K_3 .
- 5 people watch all three channels K_1, K_2, K_3 .

Find the number of people who:

- a) Watch at least one of the three news channels.
- b) Watch none of the three news channels.
- c) Watch exactly one news channel.

Exercise 6. How many positive integers n not exceeding 1000 are not divisible by any of the following three numbers: 6, 10, 15?

Exercise 7. A benefactor wants to donate 18 **identical** gift sets to 3 villages affected by natural disasters. How many ways are there to distribute them if:

- Village A cannot receive more than 6 sets ($x_A \leq 6$).
- Village B cannot receive more than 7 sets ($x_B \leq 7$).
- Village C cannot receive more than 8 sets ($x_C \leq 8$).

Exercise 8. Suppose 45% of all newspaper readers like wine, 60% like orange juice, and 55% like tea. Assume that 35% like any two of these drinks and 25% like all three.

- a) What percentage of readers like only wine?
- b) What percentage of readers like exactly two of these three drinks?

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